
Microeconomics I

Class 2 — Monotonicity, convexity, indifference curves

Pavel Kocourek

Universitat de Barcelona · Master in Economics

30 September 2026

Plan for today

10:00	Sign-up sheet; three snacks
10:10	Warm-up: two relations
10:20	Quiz results: $\geq$ vs $\gg$, $\forall$ and $\exists$
10:30	Monotonicity, local non-satiation
11:00	Break
11:10	Indifference and contour sets
11:20	Convexity, the MRS
11:35	Five preferences, in pairs
11:50	Homothetic, quasilinear; summary

Practicalities

- Group sign-up sheet going round: 2–5 names per group
- Problem set 1 out Tue 6 Oct, due in class Wed 14 Oct
- Office hours: today 12:00–13:00, after class
up to 20 minutes per student; those who book by e-mail have priority
- Slides and reading: pkocourek.com/teaching/micro-1 · pkocourek@ub.edu

Three classes, three snacks

Wed 7 Oct

?

Wed 14 Oct

?

Wed 21 Oct

?

menu: chocolate bar · crisps · nuts · biscuits · fruit · wafer

Choose all three now. Hands up: three different ones?

Set-up after Simonson (1990), study 1.

Chosen as a bundle, people mix. One at a time, they repeat.

all three chosen today

64 %

took three different snacks

one chosen each week

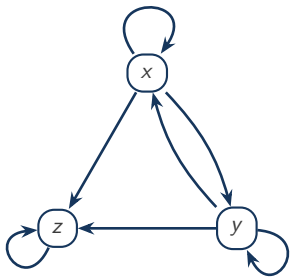
9 %

took three different snacks

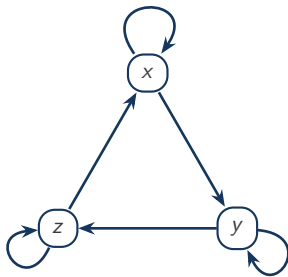
Same menu, students in a class. Which of the two choices reveals the **preference**?

Simonson (1990), *Journal of Marketing Research* 27(2), study 1. We come back to it once “convex” is defined.

In pairs: two relations on $\{x, y, z\}$



R_1



R_2

$a \rightarrow b$ means $a R b$. Complete, reflexive, transitive, symmetric, asymmetric, antisymmetric?

Four minutes. Which one is a **rational** preference?

Answers: two relations

	R_1	R_2
complete	✓	✓ every pair one way, all loops
reflexive	✓	✓ three loops
transitive	✓	✗ $x R y, y R z$, but not $x R z$
symmetric	✗ $x R z$, not $z R x$	✗ $x R y$, not $y R x$
asymmetric	✗ loops; $x R y$ and $y R x$	✗ loops
antisymmetric	✗ $x R y, y R x, x \neq y$	✓ never both ways

R_1 is rational: $x \sim y \succ z$. R_2 is complete but cycles, $x \succ y \succ z \succ x$: the money pump.

From class 1: what today builds on

$\succsim$ is **rational**: complete and transitive

MWG Def. 1.B.1

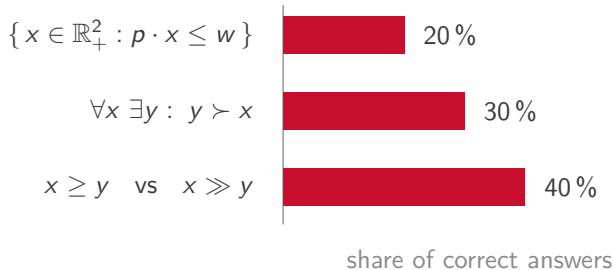
$$x \succ y \iff x \succsim y \text{ and not } y \succsim x \qquad x \sim y \iff x \succsim y \text{ and } y \succsim x$$

$\sim$ is an equivalence relation: X splits into **indifference classes**

Prop. 1.B.1

Today: what those classes look like once $\succsim$ is **monotone** and **convex**.

Diagnostic quiz: three things to fix today



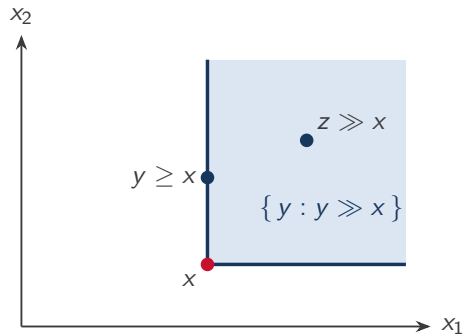
$n = 20$, mean 3.9 of 8. The ideas were fine (tangency: 95%); the **notation** was not.

Two ways to say “more”

$$y \geq x \iff y_1 \geq x_1 \text{ and } y_2 \geq x_2$$

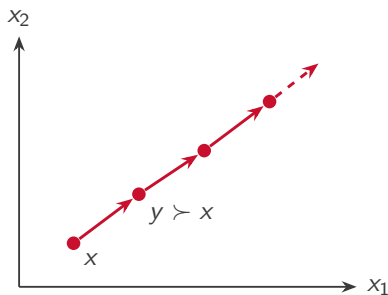
$$y \gg x \iff y_1 > x_1 \text{ and } y_2 > x_2$$

Quiz: $x = (3, 1)$, $y = (3, 2)$.



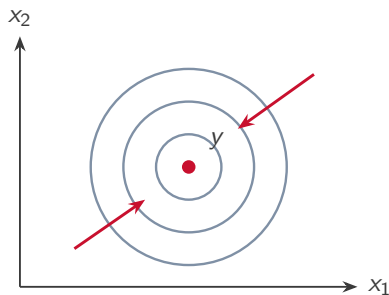
Shaded: strictly more of both. Its edges belong to $\geq$ but not to $\gg$.

Swap the quantifiers, change the claim



$$\forall x \exists y : y \succ x$$

for every bundle, a better one



$$\exists y \forall x \neq y : y \succ x$$

one bundle beats all: a bliss point

Left: no bundle is best — **non-satiation**. Its local version arrives in ten minutes.

Monotonicity

MWG 3.B

Definition: monotone, strongly monotone

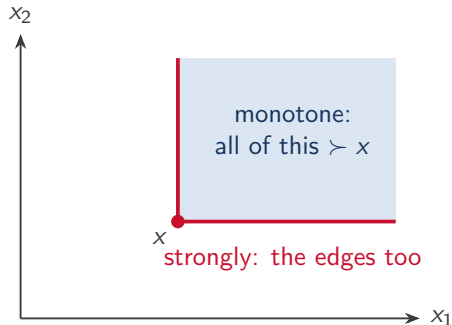
$\succsim$ on $X = \mathbb{R}_+^2$ is **monotone** if

$$y \gg x \Rightarrow y \succ x$$

strongly monotone if

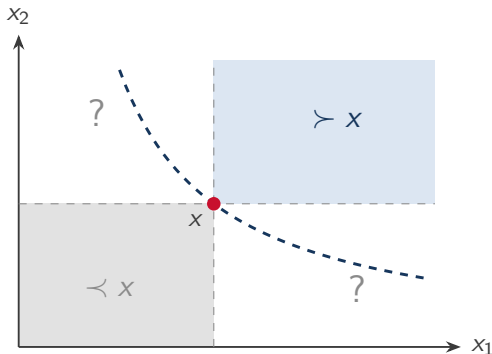
$$y \geq x \text{ and } y \neq x \Rightarrow y \succ x$$

MWG Definition 3.B.2.



Strongly monotone $\Rightarrow$ monotone. The converse fails — an example in the exercise.

Monotone: where can the indifferent bundles be?



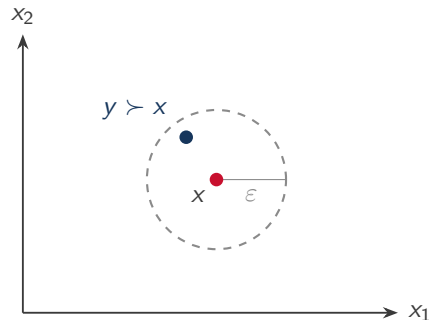
Nothing indifferent to x lies strictly north-east or strictly south-west of it. The curve through x cannot slope up.

Definition: local non-satiation

$\succsim$ is **locally non-satiated** if
for every $x \in X$ and every $\varepsilon > 0$
there is $y \in X$ with

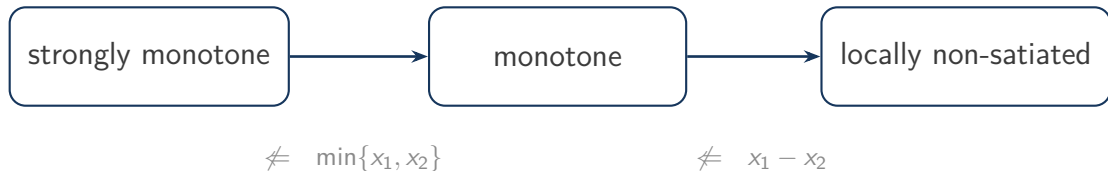
$$\|y - x\| \leq \varepsilon \quad \text{and} \quad y \succ x$$

MWG Definition 3.B.3.



$\forall x \forall \varepsilon \exists y$: the quiz's quantifier order. No direction is prescribed: y may have **less** of a good.

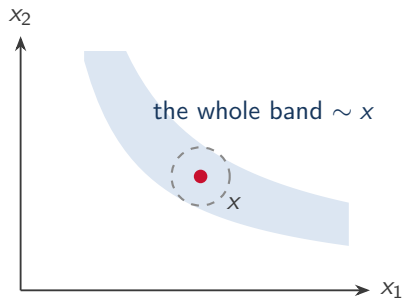
Three strengths of “more is better”



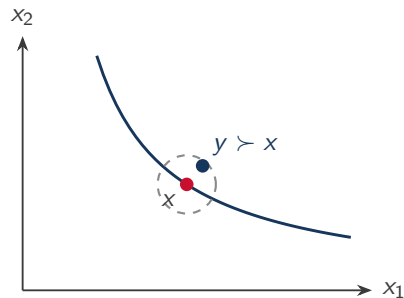
Prove monotone $\Rightarrow$ locally non-satiated in two lines. Where do you use $X = \mathbb{R}_+^2$?

Grey: a preference with the weaker property but not the stronger one, written as the number it ranks by.

Local non-satiation rules out thick indifference sets



thick: fails



thin: fine

Inside a thick band, every bundle within ε of x is indifferent to x .

After MWG Figure 3.B.2, redrawn; the circle test on the right is MWG Figure 3.B.1.

In pairs: monotone? strongly? locally non-satiated?

$$x \succsim y \iff$$

monotone

strongly monotone

LNS

$$x_1 + x_2 \geq y_1 + y_2$$

$$\min\{x_1, x_2\} \geq \min\{y_1, y_2\}$$

$$x_1 - x_2 \geq y_1 - y_2$$

$$\|x - (5, 5)\| \leq \|y - (5, 5)\|$$

$$x_1 > y_1, \text{ or } x_1 = y_1 \text{ and } x_2 \geq y_2$$

Five minutes. Give a counterexample for every $\times$.

In pairs: monotone? strongly? locally non-satiated?

$x \succsim y \iff$	monotone	strongly monotone	LNS
$x_1 + x_2 \geq y_1 + y_2$	✓	✓	✓
$\min\{x_1, x_2\} \geq \min\{y_1, y_2\}$	✓	✗	✓
$x_1 - x_2 \geq y_1 - y_2$	✗	✗	✓
$\ x - (5, 5)\ \leq \ y - (5, 5)\ $	✗	✗	✗
$x_1 > y_1$, or $x_1 = y_1$ and $x_2 \geq y_2$	✓	✓	✓

Five minutes. Give a counterexample for every ✗.

min: $(1, 1) \sim (2, 1)$. $x_1 - x_2$: $(2, 3) \prec (1, 1)$. Bliss: nothing beats $(5, 5)$.

Break

Back at 11:10

Indifference and contour sets

MWG 3.B

Three sets attached to a bundle x

$$\{y \in X : y \succsim x\}$$

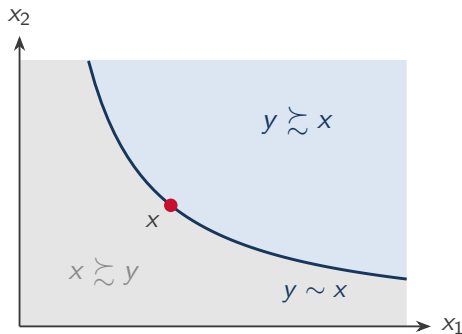
all bundles y in X such that y is at least as good as x

upper contour set $\{y \in X : y \succsim x\}$

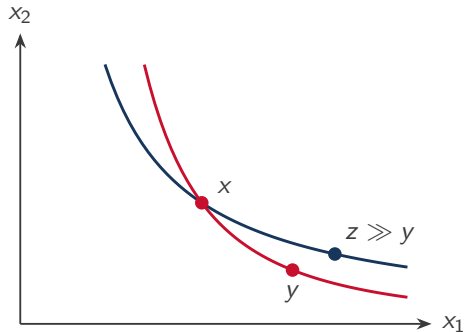
indifference set $\{y \in X : y \sim x\}$

lower contour set $\{y \in X : x \succsim y\}$

MWG 3.B, p. 43, in the text; Figure 3.B.2(b).



Indifference curves do not cross



$y \sim x$ and $x \sim z \Rightarrow y \sim z$
transitivity

$z \gg y \Rightarrow z \succ y$
monotone

Contradiction.

Rational + monotone: one indifference curve through every bundle; further out is better.

Convexity

MWG 3.B

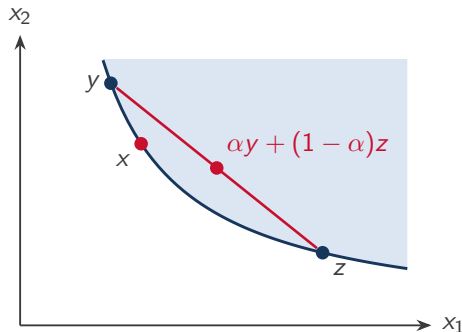
Definition: convex preferences

$\succsim$ is **convex** if every upper contour set $\{y \in X : y \succsim x\}$ is a convex set:

$$y \succsim x \text{ and } z \succsim x$$

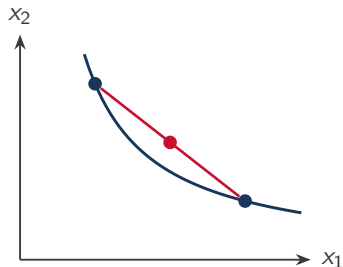
$$\Rightarrow \alpha y + (1 - \alpha)z \succsim x, \quad \alpha \in [0, 1]$$

MWG Definition 3.B.4.



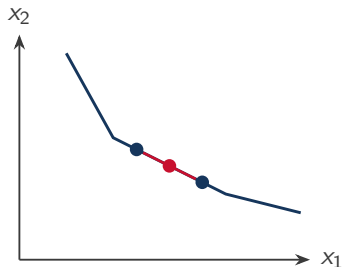
Averages are at least as good as extremes. The curve bows toward the origin.

Strictly convex, convex, not convex



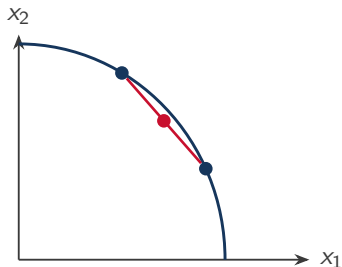
strictly convex

the mix is strictly better



convex, not strictly

a flat stretch: the mix is only $\sim$



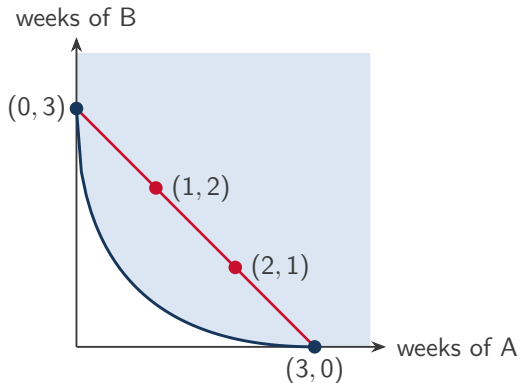
not convex

the mix is worse

strictly convex: $y \succsim x, z \succsim x, y \neq z \Rightarrow \alpha y + (1 - \alpha)z \succ x, \alpha \in (0, 1)$

MWG Definition 3.B.5. Blue curve: an indifference set; blue dots: two bundles on it; red dot: their mix.

Back to the snacks



One bundle: (weeks A, weeks B)

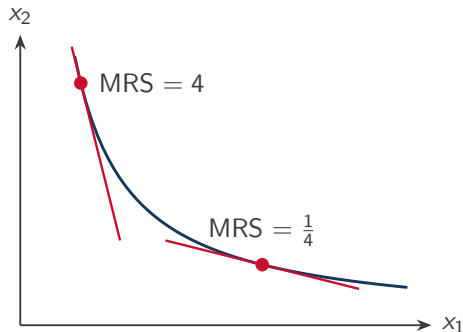
$(3, 0) \sim (0, 3)$ and convex $\succsim$
 $\Rightarrow$ every mix is $\succsim$ both

Week by week: no bundle in view

64 % mixed with the bundle in view, 9 % without. Convexity is a statement about **bundles**.

Simonson (1990), study 1. Two snacks A, B shown; the study offered six.

The marginal rate of substitution is the slope



$MRS_{12}(x)$: good 2 given up per unit of good 1, staying indifferent

$= -\text{slope of the curve at } x$

Monotone + convex: moving right along a curve, the MRS **never rises**; strictly convex: it falls.

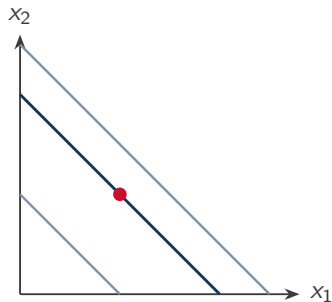
With a utility function u : $MRS_{12} = (\partial u / \partial x_1) / (\partial u / \partial x_2)$, the quiz's last question.

In pairs: five preferences on $\mathbb{R}_+^2$

	$x \succsim y \iff$	monotone	strongly	convex	strictly
perfect substitutes	$x_1 + x_2 \geq y_1 + y_2$				
perfect complements	$\min\{x_1, x_2\} \geq \min\{y_1, y_2\}$				
Cobb–Douglas type	$x_1 x_2 \geq y_1 y_2$				
quasilinear	$x_1 + \sqrt{x_2} \geq y_1 + \sqrt{y_2}$				
lexicographic	$x_1 > y_1$, or $x_1 = y_1$ and $x_2 \geq y_2$				

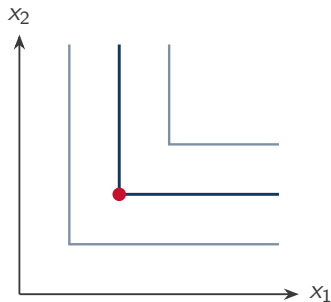
Sketch the indifference set through $(2, 2)$ for each, then fill the row. Eight minutes.

Answers: the indifference maps, I



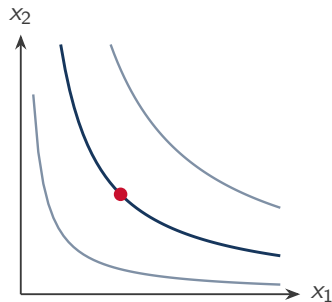
perfect substitutes

lines, slope -1



perfect complements

L-shapes, corners on the diagonal

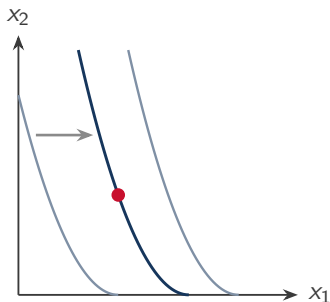


Cobb–Douglas type

hyperbolas $x_1 x_2 = c$

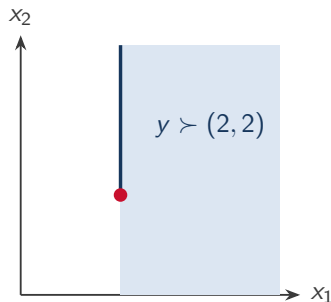
Red dot: (2, 2). Pale curves: two other indifference sets of the same preference.

Answers: the indifference maps, II



quasilinear in good 1

shifts of one curve



lexicographic

each bundle alone

Quasilinear: $x_2 = (c - x_1)^2$, $x_1 \leq c$. Lexicographic: $\{y_1 > 2\}$ plus the half-line above $(2, 2)$.

Lexicographic indifference sets are single points; Tuesday: why that matters for utility (MWG 3.C).

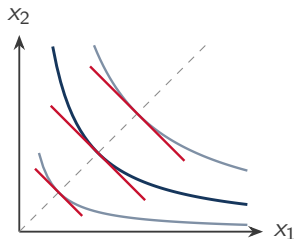
Answers: five preferences on $\mathbb{R}_+^2$

	$x \succsim y \iff$	monotone	strongly	convex	strictly
perfect substitutes	$x_1 + x_2 \geq y_1 + y_2$	✓	✓	✓	✗
perfect complements	$\min\{x_1, x_2\} \geq \min\{y_1, y_2\}$	✓	✗	✓	✗
Cobb–Douglas type	$x_1 x_2 \geq y_1 y_2$	✓	✗ ⁺	✓	✗ ⁺
quasilinear	$x_1 + \sqrt{x_2} \geq y_1 + \sqrt{y_2}$	✓	✓	✓	✓
lexicographic	$x_1 > y_1$, or $x_1 = y_1$ and $x_2 \geq y_2$	✓	✓	✓	✓

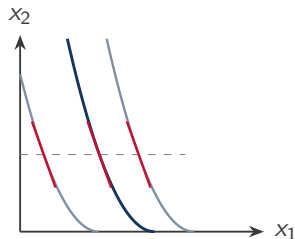
Flat stretches kill strictness: substitutes, the arms of the L.

⁺ Both fail only on the axes: $(0, 1) \sim (0, 2)$, and the mix of $(1, 0)$ and $(2, 0)$ is $\sim (0, 0)$. Off the axes, both hold.

Two families you will meet again



homothetic
same slope along a ray



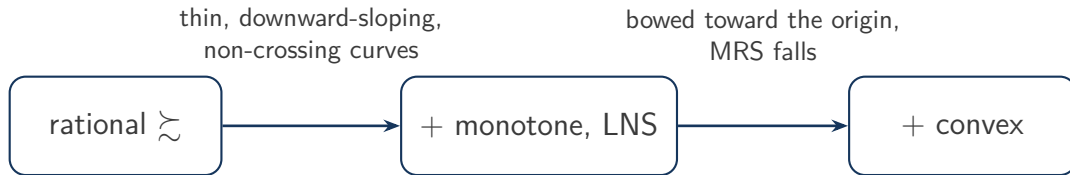
quasilinear in good 1
same slope along a horizontal

homothetic: $x \sim y \Rightarrow \alpha x \sim \alpha y, \alpha \geq 0$

quasilinear in good 1: $x \sim y \Rightarrow x + \alpha e_1 \sim y + \alpha e_1, \alpha \in \mathbb{R}; \quad x + \alpha e_1 \succ x, \alpha > 0$

MWG Def. 3.B.6 (monotone $\succsim$ on $\mathbb{R}_+^2$), Def. 3.B.7 (on $X = \mathbb{R} \times \mathbb{R}_+$, so $\alpha < 0$ stays in X); $e_1 = (1, 0)$.

Today in one line



Tuesday: when a number $u(x)$ can stand in for $\succsim$, and why the lexicographic one has none.
Problem set 1 goes out.

Read MWG 3.C (or Muñoz-García ch. 1) for Tuesday 6 October.